

```

import javax.swing.*;
import java.awt.event.*;
import java.awt.*;

class Demo extends JFrame implements ActionListener{
    char flag;
    int flag1;
    JLabel l1,l2,l3;
    JTextField t1;
    double num1,num2,value;
    JButton b1,b2,b3,b4,b5,b6,b7,b8,b9,b0,badd,bsub,bdiv,bmulti,beql,bc,bdot,bdil;

    Demo(){

        t1=new JTextField(20);
        badd=new JButton("+");
        bsub=new JButton("-");
        b1=new JButton("1");
        b2=new JButton("2");
        b3=new JButton("3");
        b4=new JButton("4");
        b5=new JButton("5");
        b6=new JButton("6");
        b7=new JButton("7");
        b8=new JButton("8");
        b9=new JButton("9");
        b0=new JButton("0");
        bdiv=new JButton("/");
        bmulti=new JButton("*");
        beql=new JButton("=");
        bc=new JButton("C");
        bdot=new JButton(".");
        bdil=new JButton("Del");

        add(t1);
        add(bc);
        add(bdil);
        add(beql);
        add(bdiv);
        add(bmulti);
        add(bsub);
        add(badd);
        add(bdot);
        add(b0);
        add(b9);
        add(b8);
        add(b7);
    }
}

```

```
add(b6);
add(b5);
add(b4);
add(b3);
add(b2);
add(b1);
b1.addActionListener(this);
b2.addActionListener(this);
b3.addActionListener(this);
b4.addActionListener(this);
b5.addActionListener(this);
b6.addActionListener(this);
b7.addActionListener(this);
b8.addActionListener(this);
b9.addActionListener(this);
b0.addActionListener(this);
badd.addActionListener(this);
bsub.addActionListener(this);
bmulti.addActionListener(this);
bdiv.addActionListener(this);
bdot.addActionListener(this);
beql.addActionListener(this);
bc.addActionListener(this);
bdil.addActionListener(this);
setVisible(true);
setDefaultCloseOperation(3);
setSize(300,400);
setLayout(null);
t1.setBounds(30,40,280,30);
b1.setBounds(40,240,50,40);
b2.setBounds(110,240,50,40);
b3.setBounds(180,240,50,40);
b4.setBounds(40,170,50,40);
b5.setBounds(110,170,50,40);
b6.setBounds(180,170,50,40);
b7.setBounds(40,100,50,40);
b8.setBounds(110,100,50,40);
b9.setBounds(180,100,50,40);
b0.setBounds(110,310,50,40);
bc.setBounds(180,380,100,40);
bdil.setBounds(60,380,100,40);
beql.setBounds(180,310,50,40);
bdot.setBounds(40,310,50,40);
bmulti.setBounds(250,170,50,40);
```

```

        bdiv.setBounds(250,100,50,40);
        bsub.setBounds(250,240,50,40);
        badd.setBounds(250,310,50,40);

    }

    public void actionPerformed(ActionEvent ae){

        if(ae.getSource()==b1){
            t1.setText(t1.getText().concat("1"));
        }
        if(ae.getSource()==b2){
            t1.setText(t1.getText().concat("2"));
        }
        if(ae.getSource()==b3){
            t1.setText(t1.getText().concat("3"));
        }
        if(ae.getSource()==b4){
            t1.setText(t1.getText().concat("4"));
        }
        if(ae.getSource()==b5){
            t1.setText(t1.getText().concat("5"));
        }
        if(ae.getSource()==b6){
            t1.setText(t1.getText().concat("6"));
        }
        if(ae.getSource()==b7){
            t1.setText(t1.getText().concat("7"));
        }
        if(ae.getSource()==b8){
            t1.setText(t1.getText().concat("8"));
        }
        if(ae.getSource()==b9){
            t1.setText(t1.getText().concat("9"));
        }
        if(ae.getSource()==b0){
            t1.setText(t1.getText().concat("0"));
        }
        if(ae.getSource()==bdot&&flag1==0){
            t1.setText(t1.getText().concat("."));
            flag1=1;
        }

    }

```

```

//num1=Float.parseFloat(t1.getText());
if(ae.getSource()==badd){

    num1=Double.parseDouble(t1.getText());
    flag='a';
    flag1=0;
    t1.setText("");
}

if(ae.getSource()==bmulti){
    num1=Double.parseDouble(t1.getText());
    flag='b';
    t1.setText("");
    flag1=0;
}

if(ae.getSource()==bdiv){
    num1=Double.parseDouble(t1.getText());
    flag='c';
    t1.setText("");
    flag1=0;
}

if(ae.getSource()==bsub){
    num1=Double.parseDouble(t1.getText());
    flag='d';
    t1.setText("");
    flag1=0;
}
if(ae.getSource()==bdil){

    t1.setText(t1.getText().substring(0,t1.getText().length()-1));
}


if(ae.getSource()==beql){
    num2=Double.parseDouble(t1.getText());
    t1.setText("");
switch(flag){
case 'a':
    value=num1+num2;
    break;
case 'b':

```

```

        value=num1*num2;
        break;
    case 'c':
        value=num1/num2;
        break;
    case 'd':
        value=num1-num2;
        break;
    default:l2.setText("567 ");
    }
    t1.setText(" "+value);
    }
    if(ae.getSource()==bc){
        t1.setText("");
    }

}
}
class Calculator{
    public static void main(String args[]){
        Demo obj=new Demo();
    }
}

```